Exp '09

Design A Simple Topology And Configure with One Router, Two Switches And PCs using CISCO Packet Tracer

Aim :-

To design and configure a simple network topology consisting of one router, two switches and multiple PCs in cisco packet Tracer, and enable communication between PCs.

definitions :-

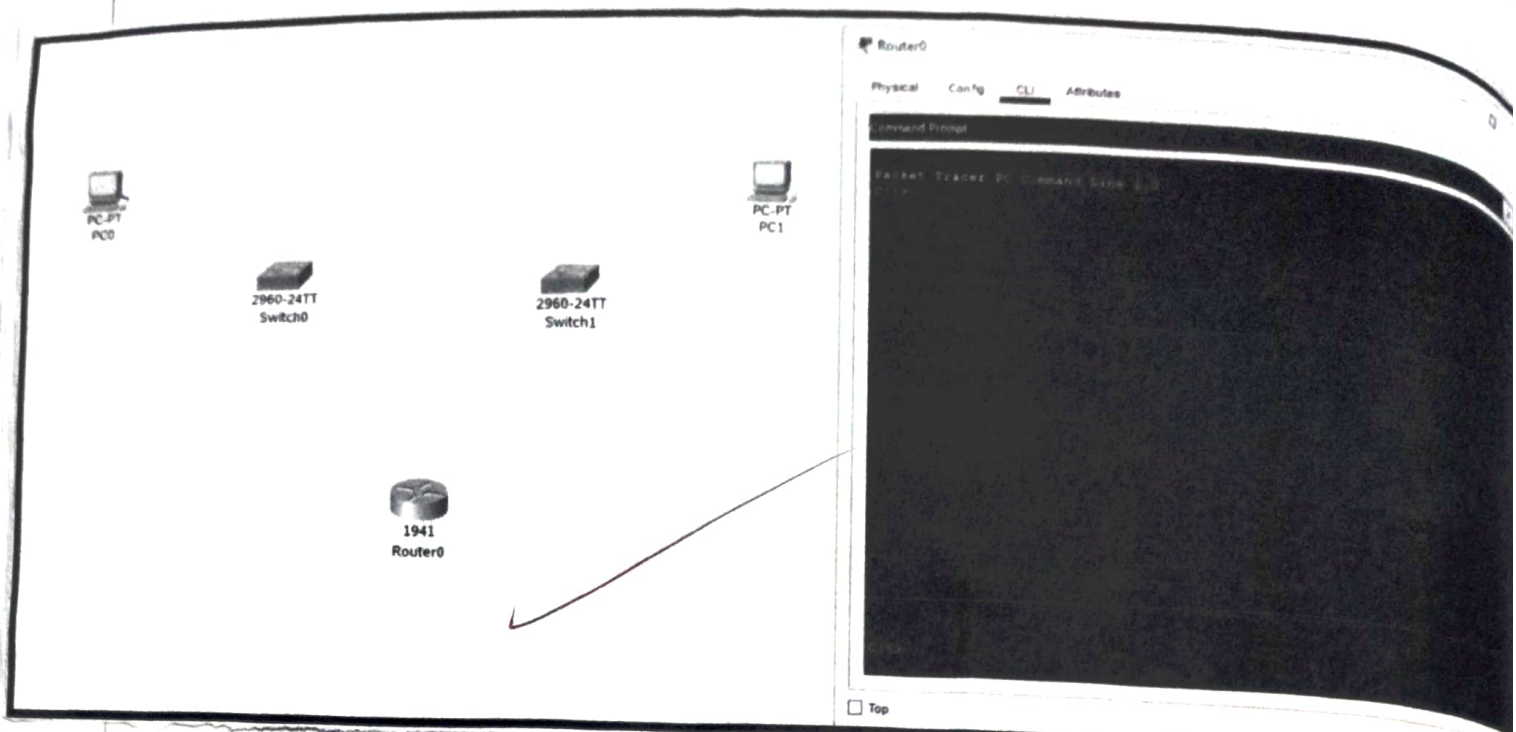
* Router * networking device that forwards data packets between computer networks, direct traffic on the internal.

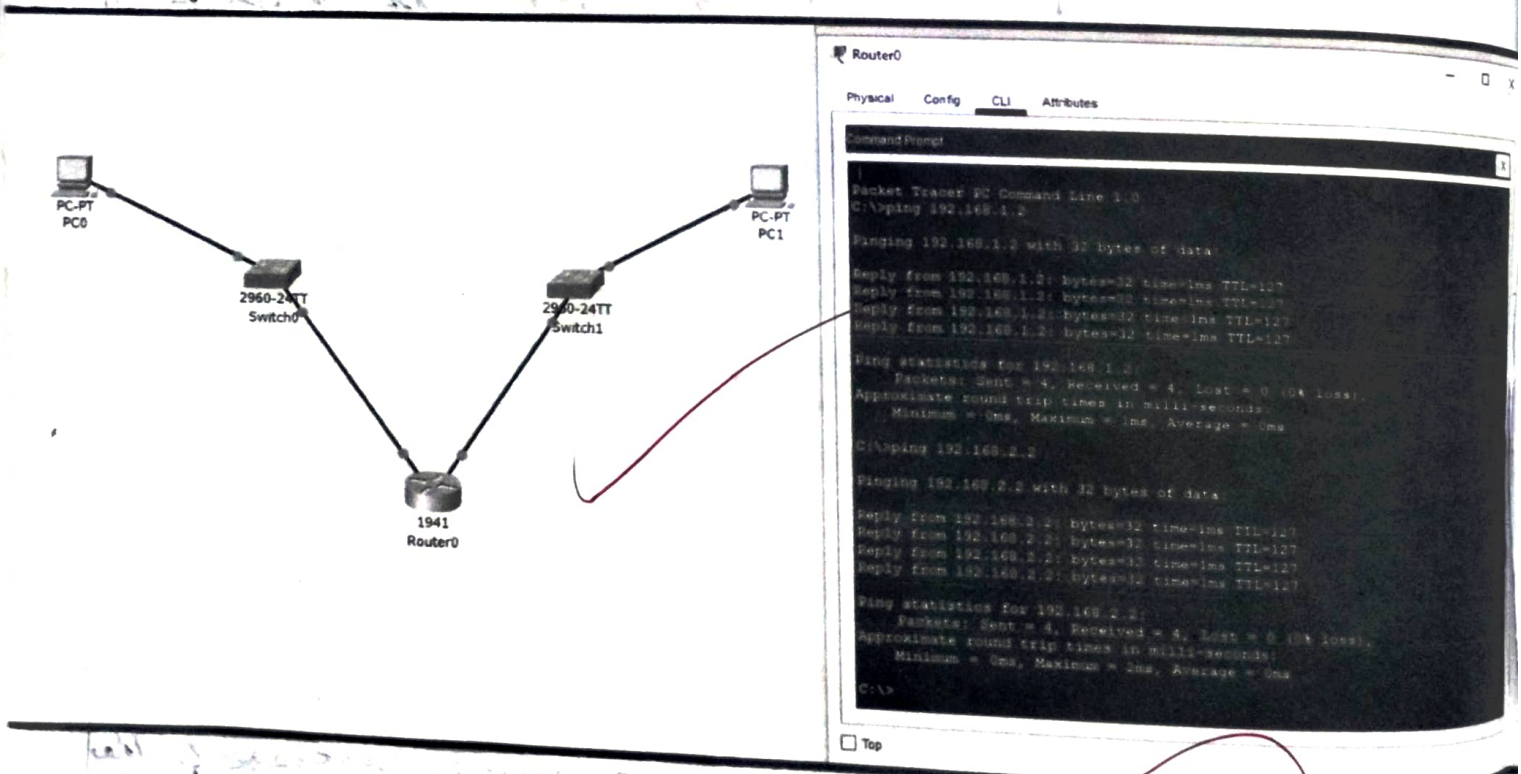
* Pc end devices in a network that user interact with send and receive data.

* straight through cable - used to connect similar devices like switch to switch.

Equipment needed :

* CISCO packet tracer software
* 1 Router
* 2 Switches
* 4 PCs
* Appropriate cables.


PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
2960-24TT
Switch1
1941
Router0
Router0
Physical
Config
CLI
Attributes
Command Prompt
Top


PC-PT
PC0
PC-PT
PC1
2960-24TT
Switch0
2960-24TT
Switch1
1941
Router0
Router0
Physical
Config
CLI
Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
Top

step - by - step procedure :

step 1 : open cisco packet tracer

* launch cisco packet tracer on your computer.

step 2 : Add service to workspace :

* from the bottom device toolbar, drag and drop

* 1 Router (PT-1)

* 2 switches (2960)

* 4 PCs

step 3 : connect devices with cables.

* uses copper straight through cables to connect.

* Router gigabit ethernet 0/0 to switch fast ethernet 0/1.

* PC3 to the switches

* PC0 to switch 0 fast ethernet 0/2

* PC1 to switch 0 Fast ethernet 0/3

* PC 2 to switch Fast ethernet 0/2

* PC3 to switch Fast ethernet 0/3

Step 4:-

* Configure Router interface
* click the router and with
* enter the following commands
* Terminal interface 142.168.1.1 255.255.255.0
shutdown exit and write memory.

Step 5:-

configure PC Ip Address

* click on each PC desktop tab → ip configuration and assign the following ip's and gateway device Ip

. p c 142.168.1.10 253. 253. 253.0 192.168.1

Step 6:-

* save and test connectivity
* save the packet Tracer file.

Result:-

The PC connected to different switches can communicate with each other via the router. The successfully graphics verify that the network is properly configured and operational.